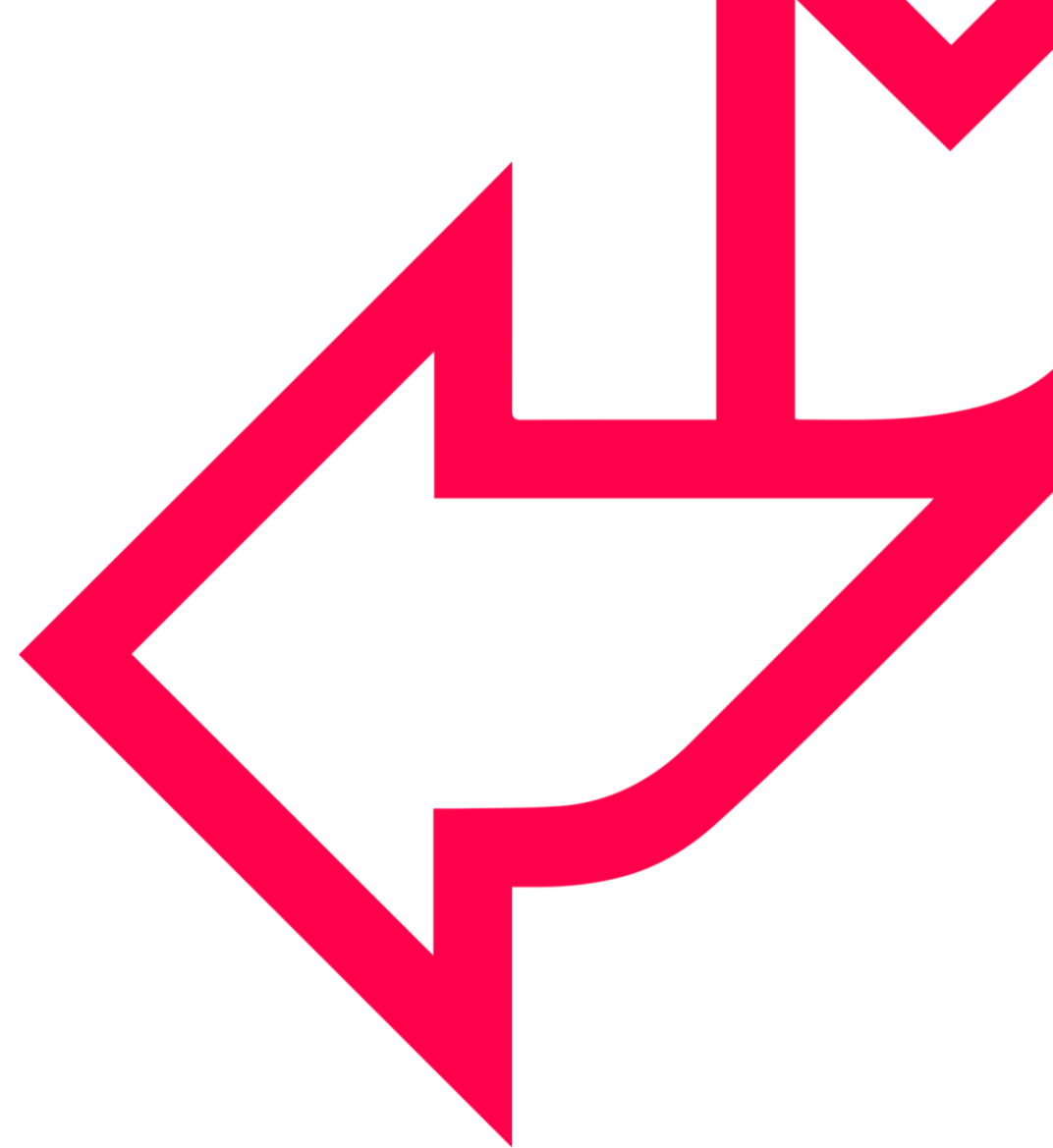




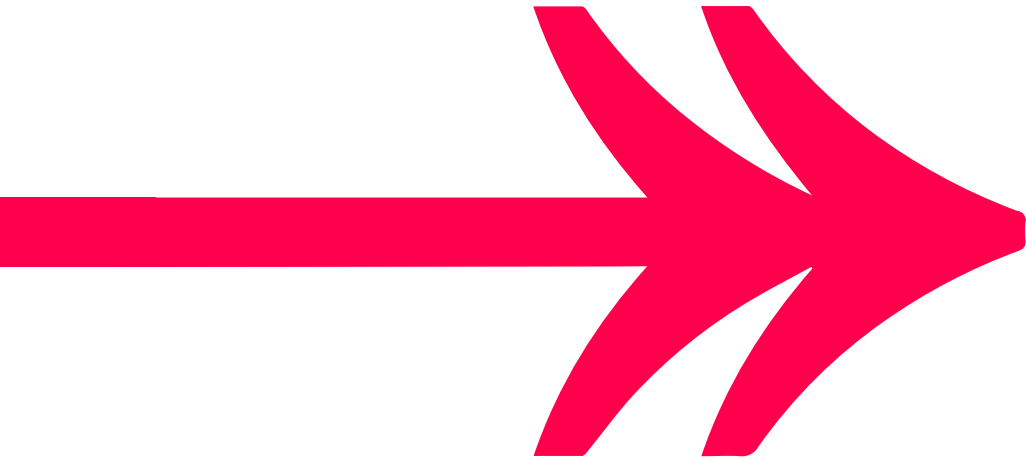
Control Flow - Iteration





Lesson Objectives

- In this chapter, you'll learn about:
- Using iteration (loops) in Python
 - while
 - For
- Using the range() function





WHAT IS CONTROL FLOW?

Control flow controls the order in which we do things

- Sequence
 - Running code step by step, in order
- Selection
 - Deciding which lines of code should run
- Iteration
 - Doing the same thing many times, i.E. In a loop





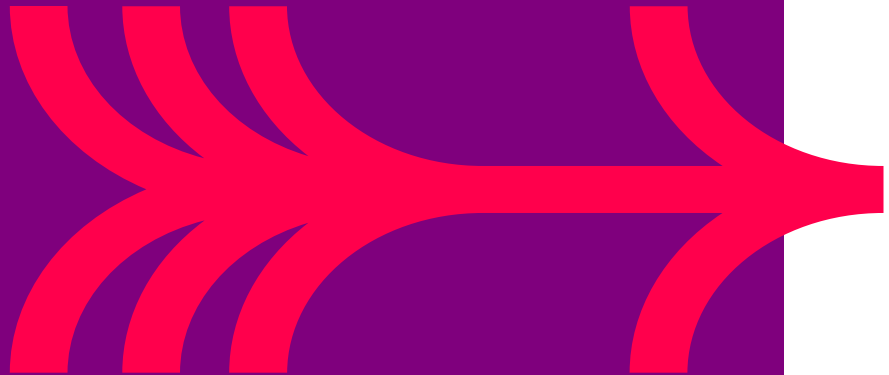
ITERATION – USING WHILE

```
x = 1
```

```
while x < 5:  
    print(x, "Hello World")  
    x = x + 1
```



```
1 Hello World  
2 Hello World  
3 Hello World  
4 Hello World  
Press any key to continue ...
```



ANOTHER WHILE EXAMPLE

```
n = 1
```

```
While n <= 5:  
    print('*' * n)  
    n = n + 1
```

```
print('*' * n)
```

```
While n > 0:  
    n = n - 1  
    print('*' *  
n)
```

```
*  
**  
***  
****  
*****
```

```
*****
```

```
*****  
****  
***  
**  
*
```

What will
n be after
the while
loop?



BREAK OUT!

Use the **break** statement to end any loop

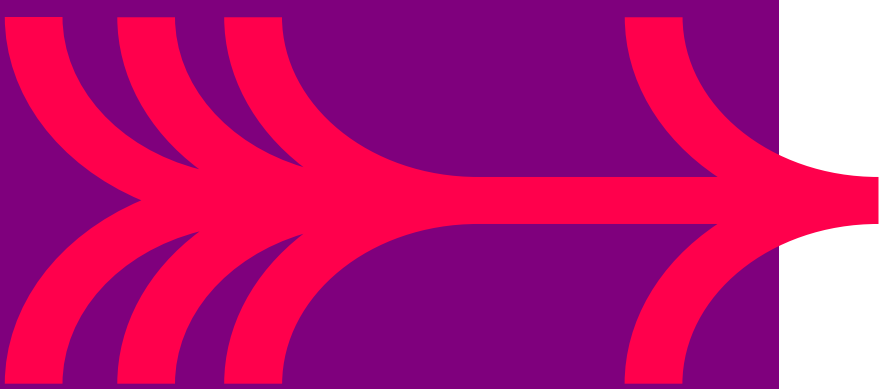
```
total = 0
answer = 'y'

while answer == 'y':
    total += int(input('Enter a number (1-10)'))

    if total >= 21:
        break

    answer = input('Get another number?')

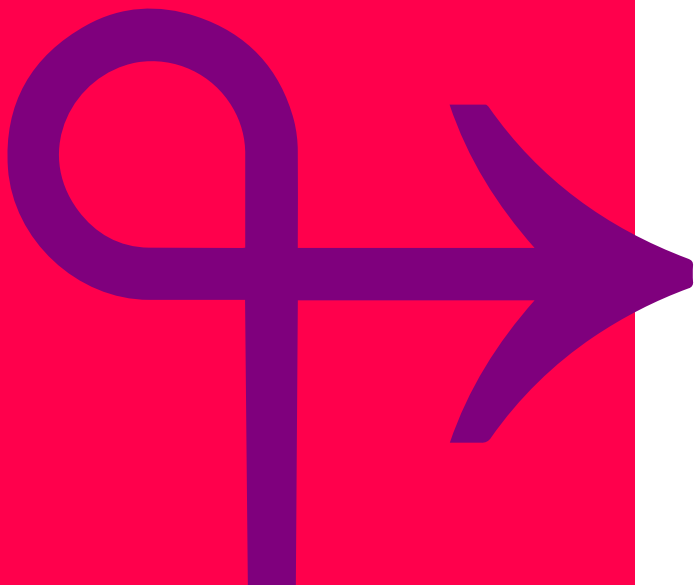
print('Total is ', total)
```





PART 2 – FOR LOOPS

- In this part you'll explore **for** loops in Python
- But before studying **for** loops, let's have a look at the very useful **range** function





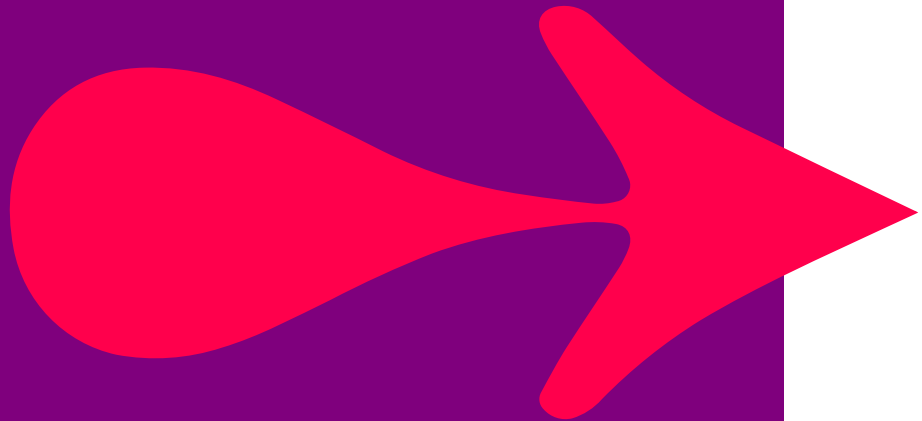
THE RANGE FUNCTION

Generate a sequence of numbers

`range(5)` 0,1,2,3,4

`range(1,5)` 1,2,3,4

`range(5,0,-1)` 5,4,3,2,1





ITERATION – FOR LOOPS

```
for x in range(5):  
    print(x, "Hello World")
```

`range(5)` means
All the numbers
from 0 to 4

```
0 Hello World  
1 Hello World  
2 Hello World  
3 Hello World  
4 Hello World
```



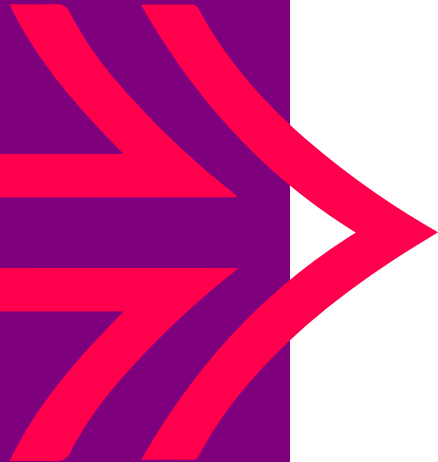


FOR LOOPS – USING RANGE

```
for x in range(2,5):  
    print(x, "Hello World")
```

All the numbers
from 2 to 4

```
2 Hello World  
3 Hello World  
4 Hello World
```

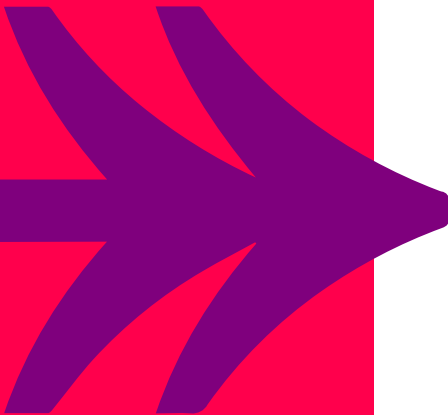


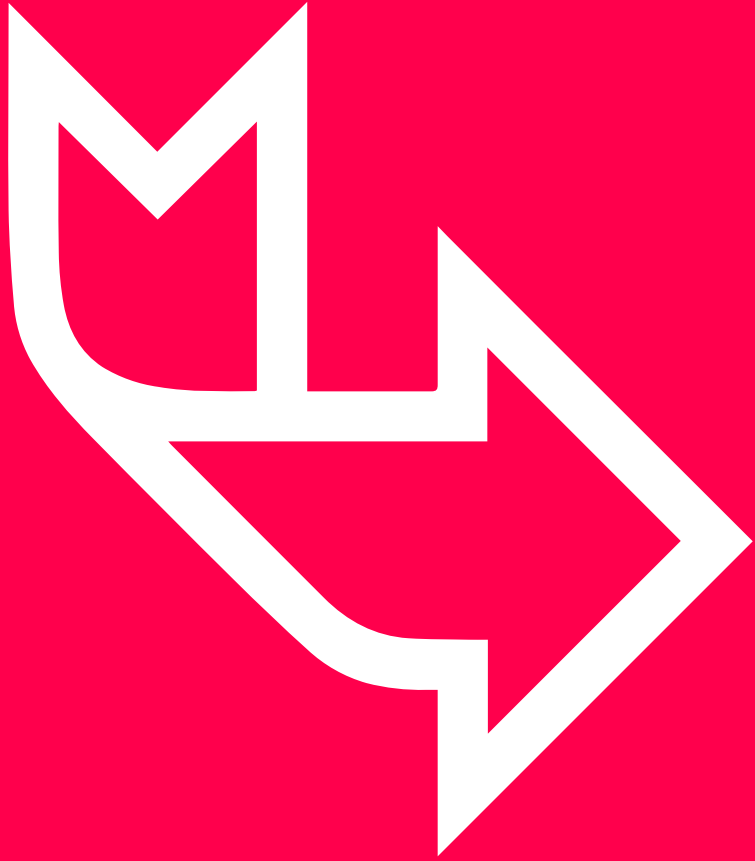


PYTHON FUNDAMENTALS

In this chapter, you learnt about:

- Using iteration (loops) in python
 - While
 - For
 - Break
- Using the range function





Guided Task

- Please open the instructions in Cloud Academy
- This guided task has many exercises
- **See how many you can do!**



FURTHER READING

- <https://www.python.org/>
- <https://www.python.org/dev/peps/pep-0008/#a-foolish-consistency-is-the-hobgoblin-of-little-minds>





Thank you!

Hope you enjoyed this learning journey

